

How to read this

Every part has a **primary** and a **backup** from a different vendor, so a single out-of-stock item cannot stall the build. Every link on this list was opened and checked on the date above. Prices are what the vendor showed that day.

Amazon is deliberately absent. Those listings rotate sellers and cannot be verified reliably, and a dead link in a parts list is worse than no link.

Two ways to buy it

Path A: all US vendors **\$132**
Adafruit, SparkFun, Pololu. Fastest and simplest.

Path B: budget **\$84**
Cheaper motors and driver, mixed vendors, slower shipping.

Plus consumables **\$19**
Both paths. Glue, tape, zip ties, brush materials.

Heads up on the budget. The design doc estimated \$72 from typical marketplace pricing. Verified vendor pricing lands higher: about \$160 all-in on Path A, about \$113 on Path B, including the rechargeable power bank added on Sep 15 (change order MOC-001, about \$35 more than AA batteries). The rest of the difference is mostly the motors and the driver board, which are the two places where buying the cheap version causes the most grief.

BRAIN AND CONTROL

01 micro:bit v2 Qty 1

Runs the whole robot. Block coding so Nolan writes the program himself, and 25 LEDs on the front so he can read sensor values on the table without a laptop plugged in.

PRIMARY **BBC micro:bit v2** · Adafruit #4781 · in stock **\$17.95**
<https://www.adafruit.com/product/4781>

BACKUP **micro:bit v2 Board** · SparkFun DEV-17287 · in stock **\$16.50**
<https://www.sparkfun.com/micro-bit-v2-board.html>

Board only. Neither listing includes a USB cable or batteries. The micro:bit v2 uses **micro-USB**, not USB-C. There is almost certainly one in a drawer already.

02 Motor driver board Qty 1

The micro:bit cannot power motors directly. This sits between them. Both options use an edge connector, so the micro:bit slides in and nothing needs soldering.

PRIMARY **SparkFun moto:bit Carrier Board (Qwiic)** · SparkFun · in stock **\$28.50**
<https://www.sparkfun.com/sparkfun-moto-bit-micro-bit-carrier-board-qwiic.html> 3 to 11V

BACKUP **Kitronik Compact Motor Driver Board** · Kitronik #5698 · in stock, ships UK to US **£10.45**
<https://kitronik.co.uk/products/5698-compact-motor-driver-board-for-the-bbc-micro-bit> ex VAT, ~\$13

Two dead ends found. Chicago Electronic Distributors comes up first in searches for this board and has stopped accepting orders entirely. PiShop's Kitronik motor driver page returns a 404. Order the Kitronik direct from Kitronik, or take the SparkFun primary and stay domestic.

This choice decides part 10. moto:bit takes power through a barrel jack, which the USB-to-barrel cable in part 10 plugs straight into. The Kitronik takes bare wires into screw terminals and would need a USB breakout instead.

03 Drive motors

Qty 2

One motor per side. Running them in opposite directions is what lets the robot spin in place, and spinning in place is how it gets away from an edge.

PRIMARY	N20 DC Motor with Magnetic Encoder, 6V 1:298 · Adafruit #4641 · in stock https://www.adafruit.com/product/4641	\$12.50 each
BACKUP	150:1 Micro Metal Gearmotor HP 6V · Pololu #997 · active · 210 RPM https://www.pololu.com/product/997	\$23.95 each
BUDGET	GA12-N20 Gear Motor with Wheel, 100 RPM 6V · Electropeak · in stock · wheel and bracket included https://electropeak.com/n20-gear-motor-with-wheel	\$2.25 each

The budget option covers parts 03, 04 and 05 in one. At \$2.25 with a wheel and a mounting bracket in the box it is a tenth the price of the primary path. The catch: their shipping page would not load, so transit time to Houston is unknown. Fine given three buffer weekends, risky if the schedule slips.

The encoder on the primary is not wasted. It counts wheel rotations, which is what a neat back-and-forth mowing pattern would need. That pattern was ruled out of Rev A for being too hard, and the encoder quietly leaves the door open for Rev B.

04 Wheels

Qty 1 pair

Press onto the motor shafts. Wheel size sets the speed, and speed is the variable the whole experiment measures, so this is not a cosmetic choice.

PRIMARY	Pololu Wheel 32×7mm Pair, black · Pololu #1087 · active https://www.pololu.com/product/1087	\$4.45 32 mm
BACKUP	Pololu Wheel 60×8mm Pair, black · Pololu #1420 · active https://www.pololu.com/product/1420	\$6.75 60 mm

Take the 32 mm. Both press-fit onto 3 mm D shafts, which is what the N20 motors have. The 60 mm wheel nearly doubles the travel speed for the same motor, which eats the stopping-distance margin the design depends on. If the 60 mm gets used, redo the speed arithmetic first.

05 Motor brackets

Qty 1 pair

Holds the motors square to the chassis. Motors glued on by eye end up pointing slightly different directions and the robot drives in a curve.

PRIMARY	Micro Metal Gearmotor Bracket Extended Pair · Pololu #1089 · active · screws and nuts included https://www.pololu.com/product/1089	\$4.95
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Skip this line if the Electropeak budget motors are ordered, since a bracket comes with each one.

06 Rear ball caster

Qty 1

The third point of contact. Two wheels and a caster is stable and turns cleanly. Four wheels would fight every turn.

PRIMARY	Pololu Ball Caster with 3/8" Metal Ball · Pololu #951 · active · height 0.4" to 0.6" https://www.pololu.com/product/951	\$2.95
BACKUP	Pololu Ball Caster with 3/8" Plastic Ball · Pololu #950 · same height range, lighter https://www.pololu.com/product-info-merged/950	~\$2.50

Use every spacer in the bag. A 32 mm wheel puts the axle 0.63" off the table. The caster tops out at 0.60" with all its spacers fitted, which leaves the robot tipped forward by about a millimeter. That is fine, and it is the correct direction to be off, because it points the sensors slightly down. Build it with no spacers and the robot sits nose-up and the sensors read wrong.

07 Cliff sensors

Qty 2

The heart of the project. Each one shines infrared down at the table and measures the bounce. Table there, light comes back. Table gone, nothing comes back. Two of them, one at each front corner, because a corner arrives at one side well before the other.

PRIMARY

QTR-1A Reflectance Sensor, 2-pack · Pololu #2458 · active · analog out, 5V

<https://www.pololu.com/product/2458>

\$5.65

2-pack

BACKUP

TCRT5000 IR Reflective Optical Sensor, 2-pack · ProtoSupplies · 46 in stock · bare sensor

<https://protosupplies.com/product/tcrt5000-ir-reflective-optical-sensor/>

\$0.49

2-pack

Mount them 3 mm above the table. The QTR-1A is sharpest at 3 mm and unreliable past 6 mm, which is tighter than it sounds. Build the sensor boom so the height can be shimmed, and check it with a ruler before every test run.

The backup is cheap because it is not finished. The ProtoSupplies part is the raw sensor with no circuit around it. It needs a 150 ohm resistor on the emitter and a 4.7k on the receiver, wired by hand. A fine afternoon of learning, a bad surprise the weekend the sensors were supposed to just work.

08 Backup whisker switch

Qty 1

A stiff wire dragging on the table holding a switch closed. When the table ends the whisker drops and the switch opens. It cannot be fooled by a dark placemat or a shiny finish, because it is not looking at anything, it is touching.

PRIMARY

Micro Switch with Roller Lever, three terminal · Adafruit #819 · in stock

<https://www.adafruit.com/product/819>

\$1.95

Two methods, one job. Optical and mechanical fail in completely different situations, which is exactly why both are on the robot. Real safety systems are built this way, and it is the best answer Nolan has for a judge asking what happens when the sensor is wrong.

CLEANING AND POWER

09 Sweeper brush motor

Qty 1

Spins the brush that flicks crumbs up and backward into the tray. It only ever runs one direction at one speed, so it wires straight to the battery through its own small switch instead of needing a third driver channel.

PRIMARY

DC Toy / Hobby Motor, 130 size · Adafruit #711 · in stock · 4.5 to 9V

<https://www.adafruit.com/product/711>

\$1.95

Current check. 70 mA unloaded, 250 mA loaded, 500 mA if the brush jams on something. The power bank supplies up to 2.4 A, so a jammed brush does not brown out the micro:bit.

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Rechargeable power bank and cable

Changed Sep 15

Replaces the 4 x AA holder (change order MOC-001). The bank rides in a printed sleeve and charges through USB-C without coming out, and it holds a steady 5 V, so the speed test is not skewed by fading batteries. It is a sealed commercial pack with its own protection circuits. Links checked Sep 15, 2026.

PRIMARY

Anker 321 Power Bank (PowerCore 5K), A1112 · Walmart · 97 x 45.8 x 22 mm, 5,200 mAh, USB-C in, USB-A out
<https://www.walmart.com/ip/Anker-5K-Power-Bank-12W-USB-C-USB-A-Port/2335161189>

\$23.84

BACKUP

Same bank · Target A-86373147
<https://www.target.com/p/anker-5000mah-power-bank-powercore-5k-12w-1a1c-black/-/A-86373147>

~\$20

PRIMARY

USB to 2.1mm Male Barrel Jack Cable, 22AWG, 1 m · Adafruit #2697 · in stock
<https://www.adafruit.com/product/2697>

\$2.75

PRIMARY

In-line power switch for 2.1mm barrel jack, 2 A · Adafruit #1125 · in stock
<https://www.adafruit.com/product/1125>

\$2.50

PRIMARY

Right-angle 5.5 x 2.1 mm barrel adapter, male to female, 10 pack · Walmart 379576222 · added Sep 27 by MOC-006 · 30 x 24 x 11 mm
<https://www.walmart.com/ip/DC-Power-Connector-90-Degree-Right-Angle-5-5mm-x-2-1mm-Male-to-Female-10Pcs/379576222>

\$12.49

PRIMARY

Cable Matters 90 degree USB-A adapters, left and right pair · Amazon B07L6RW9RT, the one exception to the no-Amazon rule · backup Walmart 709931832
<https://www.amazon.com/Cable-Matters-Combo-Pack-Degree-Adapters/dp/B07L6RW9RT>

~\$10

The adapter is not optional. A straight USB plug sticks out about 35 mm from the side of Dusty and pushes it past the 5 inch limit. The 90 degree adapter turns the cable forward along the side. Keep whichever of the pair points forward.

The barrel adapter is not optional either. The moto:bit power socket points out the left side. A straight plug sticks out about 20 mm past the wheels. The L adapter turns the cable forward along the deck edge.

No trickle mode. Two presses of the bank button turn on trickle mode, which caps the output under 0.5 A and shuts off after 2 hours. Dusty needs more than that. One press to wake it is all it needs.

OPTIONAL, REV B ONLY

11

Laser distance sensor

Optional

Measures the real distance to the table in millimeters using a laser, so a black placemat and white paper look identical to it. This is the correct engineering answer to the dark-surface problem and a strong "what I would do next" slide.

OPTIONAL

Adafruit VL53LOX Time of Flight Distance Sensor · Adafruit #3317 · in stock
<https://www.adafruit.com/product/3317>

\$14.95

It cannot sit where the other sensors sit. Its minimum range is about 30 mm, so mounted 3 mm above the table it reads nothing but garbage. It has to go on a small mast roughly 40 mm up, pointing straight down: table present reads about 40 mm, table gone reads the floor at 700 mm or more. That is an unmistakable difference, but it changes the front end of the robot, so it is a Rev B job and not a drop-in swap.

NOT ELECTRONICS, BUY LOCALLY

ITEM	WHAT IT IS FOR	EST.
Foam board sheet	Rev A chassis. Cuts in seconds, which is the whole point of Rev A.	\$4
Shallow plastic food container	Rev B chassis. Flat floor, straight walls, and the lid becomes the debris tray.	\$4
Hot glue sticks	Mounting everything.	\$4
Zip ties, small	Cable management. Loose wires are what snag the brush.	\$4
Frost King R338H rubber foam tape (Home Depot)	Shims that keep the power bank from rattling in its sleeve. Sticky side on the sleeve, never on the bank.	\$3
Scotch-Mount double-sided mounting tape	Sticks the in-line switch to the deck. Too strong for the power bank itself.	\$5
USB-C phone charger	Charging the power bank inside Dusty. Check the drawer first.	\$0
Paintbrush, toothbrushes, pipe cleaners, felt	Brush material. Buy three kinds and test them, that is a data point.	\$6
Micro-USB cable	Programming the micro:bit. Check the drawer first.	\$0

ORDER CHECKS BEFORE CLICKING BUY

Match the power cable to the board

Barrel plug for moto:bit, bare wires for Kitronik. The single easiest way to end up with a part that does not fit.

Do not buy an L298N

It is in half the tutorials online and it throws away about 2 volts as heat. Harmless on a 12V robot, crippling on a 5V power bank.

32 mm wheels, not 60 mm

The bigger wheel nearly doubles the speed, and the stopping-distance margin is exactly what this project measures.

Buy one spare motor

At \$2.25 on the budget path there is no reason not to. One will fail, and it will fail the weekend before it is due.

The micro:bit is micro-USB

Not USB-C. Worth confirming before the first weekend rather than discovering it that morning.

Order everything in one go

Three vendors arriving on three different days costs a whole weekend. One order per vendor, all placed the same day.